Design Guide for Dubai:

Minimum design standards for mechanical services



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The Chartered Institution of Building Services Engineers
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Note from the publisher

This publication is primarily intended to provide guidance to those responsible for the design, installation, commissioning, operation and maintenance of building services. It is not intended to be exhaustive or definitive and it will be necessary for users of guidance to exercise their own professional judgement when deciding whether to abide by or depart from it.

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This document aims at establishing the minimum design standards for the design of HVAC/mechanical systems for buildings in Dubai. It does not relieve the designer of responsibility for accurately determining design arrangements and/or complying with the local authority requirements and/or relevant international codes or regulatory requirements. It should be used for initial design guidance, and provides minimum design requirements for specific design issues.

This guide has been produced in good faith by the CIBSE UAE Technical Committee to promote best engineering practices in Dubai, without legal or financial liability of CIBSE UAE or the authors, in order to assist engineers coming from elsewhere in the world to work in Dubai, UAE. It should be read alongside existing CIBSE guidance relevant to the region, including Buildings for Extreme Environments: *Arid*.

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1 Design codes, standards and guidance

The design, specification and installation of mechanical services should be in accordance with good engineering practice and should conform to local codes, standards and guidelines, and follow applicable international codes, standards and guidelines where indicated.

1.1 Local codes, standards and guidance

Source/Reference	Code, regulation, guideline reference
Dubai Municipality	*Al Sa'fat Dubai Green Building Evaluation System*
Dubai Municipality	Circulars
Dubai Civil Defence	*UAE Fire & Life Safety Code of Practice*
ESMA UAE.S 5010-1 – Latest	*Labeling – Energy efficiency label for electrical appliances* Part 1: Household air conditioners
Dubai Civil Defence	Circulars

1.2 International codes, standards and guidance

Source/Reference	Code, regulation, guideline reference
ASHRAE Standard 55 – Latest	*Thermal environmental conditions for human occupancy*
ASHRAE Standard 62.1/62.2 – Latest	*Ventilation for acceptable indoor air quality*
ASHRAE Standard 90.1 – Latest	*Energy standard for buildings except low-rise residential buildings*
ASHRAE Standard 170 – Latest	*Ventilation for health care facilities*
CIBSE	Guidance
CIBSE Commissioning Code A	*Air distribution systems* (2006)
CIBSE Commissioning Code W	*Water distribution systems* (2010)
CIBSE Commissioning Code L	*Lighting* (2018)
CIBSE Commissioning Code C	*Automatic controls* (2001)
CIBSE Commissioning Code R	*Refrigeration systems* (2002)
CIBSE Commissioning Code B	*Boilers* (2002)
SMACNA	Sheet Metal Air Conditioning Contractors National Association
BESA DW144	*Specification for sheet metal ductwork*
BESA DW172	*Specification for kitchen ventilation systems*
BS 5422 – Latest	*Method for specifying thermal insulating materials for pipes, tanks, vessels, ductwork and equipment operating within the temperature range -40 °C to +700 °C*
BS 8233 – Latest	*Guidance on sound insulation and noise reduction for buildings*
Department for Education Building Bulletin 93	*Acoustic Design of Schools – A Design Guide (UK)*

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1.3 Minimum building fabric parameters

Building fabric details	Thermal performance – (considered in heat load calculation as per DGBR requirement) (W/m².K)	Thermal performance – (recommended as per al sa'fat minimum requirements) (W/m².K)
external wall/cladding	U-value: 0.57/0.4 (circular 214)	wall (gold, platinum) – 0.42 wall (bronze, silver) – 0.57
windows – glass area 40–60% of external wall area	U-value: 1.9 (max) shading coefficient: 0.32 (max) light transmittance: 0.1 (min)	bronze, silver: 1.9 (max) shading coefficient: 0.32 (max) light transmittance: 0.1 (min) gold, platinum: 1.9 (max) shading coefficient: 0.25 (max) light transmittance: 0.1 (min)
windows – glass area > 60% of external wall area	U-value: 1.9 (max) shading coefficient: 0.25 (max) light transmittance: 0.10 (min)	bronze, silver: 1.9 (max) shading coefficient: 0.25 (max) light transmittance: 0.1 (min) gold, platinum: 1.7 (max) shading coefficient: 0.25 (max) light transmittance: 0.1 (min)
windows – glass area < 40% of external wall area	U-value: 2.1 (max) shading coefficient: 0.40 (max) light transmittance: 0.25 (min)	bronze, silver: 2.1 (max) shading coefficient: 0.4 (max) light transmittance: 0.25 (min) gold, platinum: 1.9 (max) shading coefficient: 0.32 (max) light transmittance: 0.25 (min)
skylights – glazing portion > 10% of total roof area	U-value: 1.9 (max) shading coefficient: 0.25 (max) light transmittance: 0.3 (min)	1.9 (max) shading coefficient: 0.25 (max) light transmittance: 0.3 (min)
skylights – glazing portion </=10% of total roof area	U-value: 1.9 (max) shading coefficient: 0.32 (max) light transmittance: 0.4 (min)	1.9 (max) shading coefficient: 0.32 (max) light transmittance: 0.4 (min)
roof	U-value: 0.3	bronze, silver: 0.3 gold, platinum: 0.3
exposed floors	—	gold, platinum: 0.42 bronze, silver: 0.57
Notes: The Al Sa'fat requirements for glazing are based on the assumption that the glazing ratio (total area of external walls that let in light to the total area of external envelope) is above 60%. If the final ratio will be lower, the requirements can be relaxed in line with Al Sa'fat requirements.		

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1.4 External design temperatures

Parameter	Value	Data source
dry bulb design temperature	46ºC	*Al Sa'fat Dubai Green Building Evaluation System* – 501.03 Air conditioning design parameters
wet bulb design temperature	29 ºC	*Al Sa'fat Dubai Green Building Evaluation System* – 501.03 Air conditioning design parameters
winter dry bulb	12.9 ºC	ASHRAE Handbook: *Fundamentals* Chapter 14 – Climatic design information
The building fabric and fresh air loads should be sized based on 46 ºC/29 ºC and 34 ºC/32 ºC respectively as per *Al Sa'fat* – 501.03.		

1.5 Cooling load calculations

The building cooling load calculations should be undertaken and documented utilising software approved by the Dubai Municipality.

1.6 Thermal loads

People density should be based on the architectural furniture arrangement, and where this is not available should then be based on *Al Sa'fat* – 401.01 or latest ASHRAE standards 62.1, 62.2 and 170. The associated sensible and latent loads should be based on ASHRAE Handbook – *Fundamentals*.

Lighting power density should be based on *Al Sa'fat* requirements as shown below.

1.7 Lighting power density: interior

For new buildings, the average lighting power density for the interior connected lighting load must not exceed the values below.

Building type	Maximum average (W/m²) across total building area
commercial/public: offices, hotels, resorts, restaurants	10
educational facilities	12
manufacturing facilities	13
retail outlets, shopping malls, workshops	14
warehouses	8
Lighting power density values for the building types not listed should not exceed the values indicated in the latest edition of ASHRAE Standards 90.1 and 90.2 or equivalent as approved by the Dubai Electricity and Water Authority (DEWA).	

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1.8 Lighting power density: exterior

For all new buildings, the average lighting power density for the exterior connected lighting load for specific building types must not exceed the values given below.

Building area	Maximum (W/m²) or linear metre
uncovered parking lots and drives	1.6 W/m²
walkways < 3 m wide	3.3 W/linear metre
walkways >/= 3 m wide	2.2 W/m²
outdoor stairways	10.8 W/m²
main entries	98 W/linear metre of door width
other doors	66 W/linear metre of door width
open sales areas (including vehicle sales lots)	5.4 W/m²
building façades	2.2 W/m² for each illuminated wall or surface or 16.4 W/linear metre for each illuminated wall or surface length
entrances and gatehouse inspection stations at guarded facilities	13.5 W/m²
drive-up windows at fast food restaurants	400 W/drive-through
Lighting power density values for exterior areas for building types not listed should not exceed the values indicated in the latest edition of ASHRAE Standard 90.1 or equivalent as approved by DEWA.	
If the light power density values for external lighting exceed the values specified, the additional lighting load must be powered entirely through renewable energy sources such as photovoltaic systems or similar.	

1.9 Infiltration rates

The design should take into account infiltration due to windows and door frames as per *Al Sa'fat* – 501.05 with a maximum leakage rate of 10 m³/hr/m² of building envelope at ΔP of 50 Pa. As a minimum, an infiltration rate of 0.15 air changes/hour is recommended to be used in the thermal modelling exercise in order to predict the building cooling loads. However for airside equipment sizing, we recommend using 0.25 ACH for the areas near façades/external walls and 0.5 ACH for doors opening to the outside or higher depending on the door opening size.

1.10 Ventilation rates

The ventilation rates should generally be based on the occupancy density and in accordance to *Al Sa'fat* – 401.01, ASHRAE 62.1, 62.2 and 170.

1.11 Diversity and safety factor

The building calculations should also account for diversity profiles for people, lighting, equipment, etc., and should be considered in the air conditioning calculations. Sensible and latent safety factors of 10% and 5% respectively as identified in *Al Sa'fat* – 501.03 should be used in calculating heat load.

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1.12 Building pressurisation

The air conditioning and ventilation systems should be designed such that the building will be maintained at a positive pressure. Generally the building should be pressurised with respect to the outside by approximately 10% to assist with limiting infiltration of unconditioned ambient air.

1.13 Air movement

The air movement within the occupied zone should be designed to be in accordance with CIBSE recommendations to prevent draughts and the associated feelings of discomfort, and should be within the prescribed limits of 0.20 m/s to 0.30 m/s as per *Al Sa'fat* – 402.01.

1.14 Noise and vibration control

Building services should be designed and installed as per the 'Sound and vibration control' section of the ASHRAE *Handbook*, CIBSE Guide B4: *Noise control*, CIBSE Guide A: *Environmental design* and the advice of an acoustic consultant employed on the project. The acoustic performance relating to internal noise criteria from various sources should meet *Al Sa'fat* – 403.01 requirements.

1.15 Energy conservation

Systems should be designed to minimise energy use, and *Al Sa'fat* – 502.01 should be used to drive this.

Energy conservation methods should be considered for all mechanical systems, in particular fans and pumps, which should be provided with variable speed drives whether they are constant volume systems or not.

For air handling units (AHUs) that comprise supply and extract in one unit, consideration should be given to energy recovery systems. Thermal wheel, plate heat exchangers and heat pipes should be considered for recovery of heat and the relative merits, including a quantitative assessment of efficiencies, and should be in accordance with *Al Sa'fat* – 502.10.

Demand control ventilation is to be applied in all areas that have a maximum occupancy density of 25 people or more per 100 m^2, for spaces larger than 100 m^2. Occupancy density from architectural furniture arrangement should be considered, and where this is not available this should then be based on *Al Sa'fat* – 401.01 or latest ASHRAE standards 62.1, 62.2 and 170. Demand control ventilation can include the installation of carbon dioxide (CO_2) sensors, which should control and minimise the amount of fresh air supplied to the space.

Effective energy management requires close and consistent control of all energy-consuming systems and components.

1.16 Refrigerants

All refrigerants should be environmentally friendly with zero ozone depletion potential or with global warming potential < 100, as highlighted in *Al Sa'fat* – 701.05.

1.17 Cooling towers

Cooling towers should be selected to achieve the minimum energy performance requirements of the latest version of ASHRAE 90.1. All cooling towers should be CTI certified and should be selected at 30.5 °C wet bulb.

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1.18 Chiller coefficient of performance (CoP)

Chiller CoP should be in accordance with *Al Sa'fat* – Reference Table 502.01 (1).

1.19 Hydraulic pressures and temperature

Water pipework and ductwork system operating/working and test pressures and temperatures should be determined in accordance with the design parameters set by the mode of operation of that particular system and in full compliance with the appropriate specification, British Standard, code of practice or CIBSE guide and all relevant regulations.

1.20 Fire and life safety systems

The ventilation systems associated with fire and life safety (FLS) should be informed by the FLS strategy for the building. However for basement car parking, the extract rate should be 6 ACH for normal mode and 10 ACH for smoke mode, with the necessary air make-up evenly distributed over the car parking area. The system should be designed in accordance with the *UAE Fire and Life Safety Code of Practice* (DCD, 2018) corridor smoke management system and staircase pressurisation system.

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